Plotting Tool

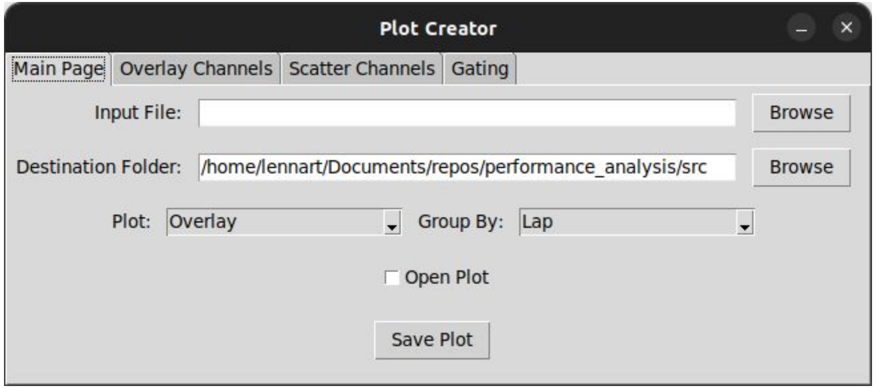
for quick data visualisation

Overview

This is a guide on how to use the plotting tool.

The plotting tool allows to quickly check unified data by providing an interface for generating standardised line and scatter plots.

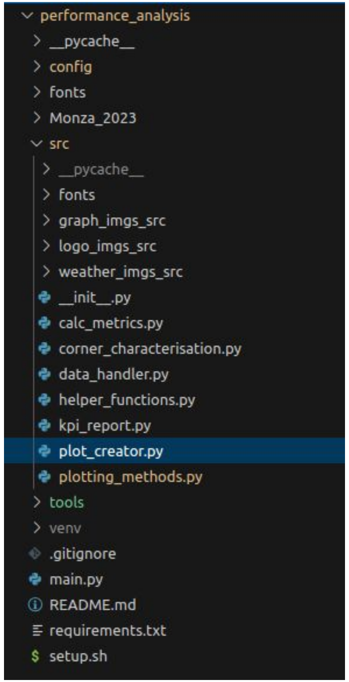
Plotting channels can be selected using a simple GUI. The functionality is explained in the following slides.



Start Plotting Tool

There are two ways to start the plotting creator:

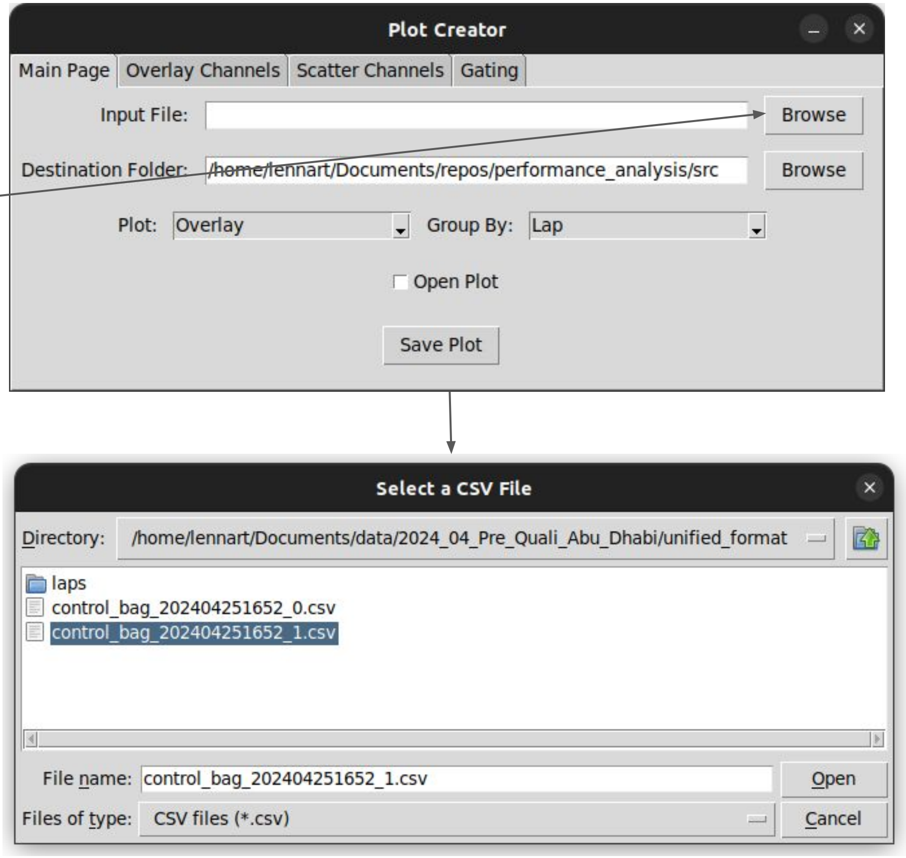
1. Manually execute python script (src/plot_creator.py).



2. After executing the main.py prompt will open for starting the plot creator.

Select Data

1. Click on Browse button next to Input File field to select input data.
2. This will open a window which allows you to select .csv files. The .csv needs to be in the format where each column is associated to a data signal with the first row holding the signals name.



Plot Options

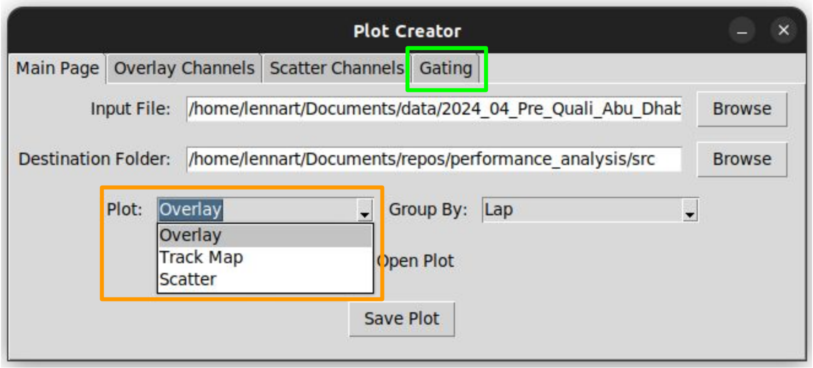
There are three plot options available.

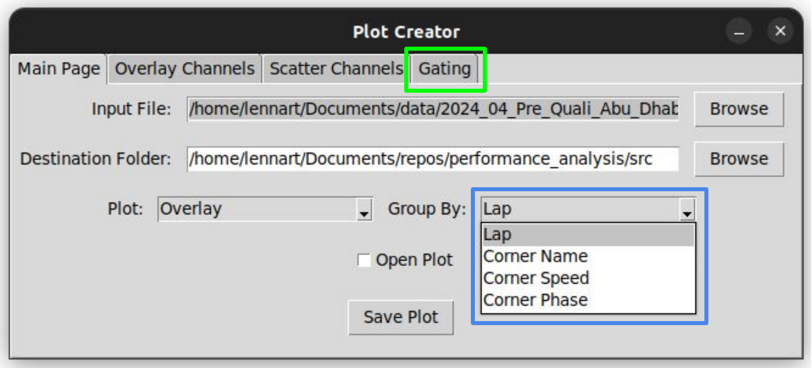
1. Overlay
2. Track Map
3. Scatter

The Group By selector allows to analyse and compare data at different abstraction levels:

1. per Lap
2. per Corner Name
3. per Corner Speed
4. per Corner Phase

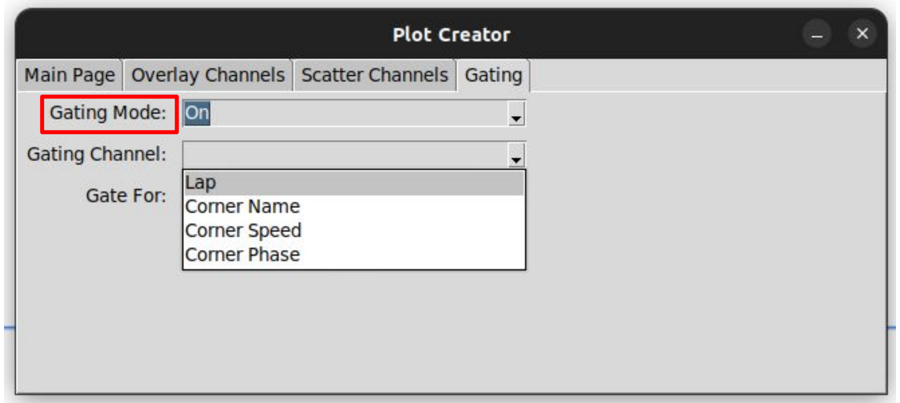
Per default the plot will created from all data stored in the .csv. If there is the need to look at something specifically there are options for gating the data in the respective tab.

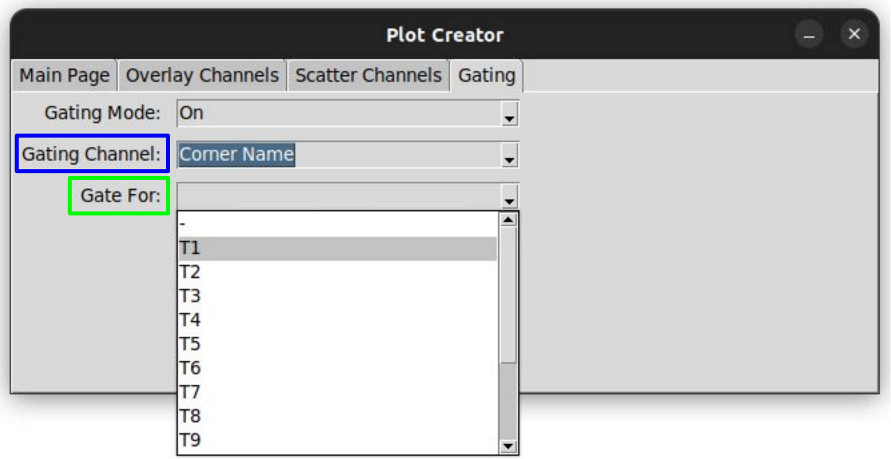




Gating

1. Turn on Gating Mode if required.
2. Select Gating Channel. There are four options available.
3. Based on gating channel selection the Gate For dropdown will show the gating options. For example all available corners will be selectable.

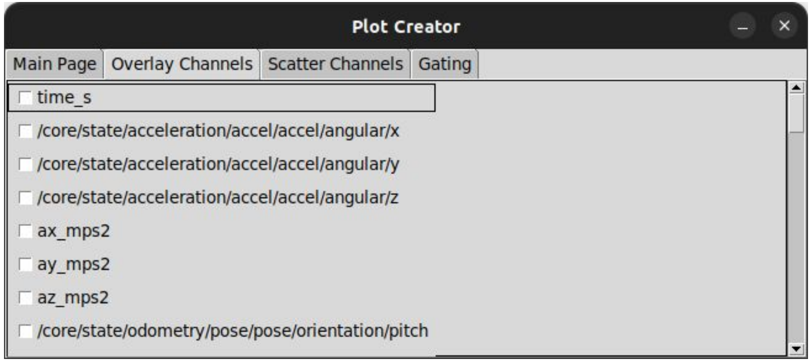


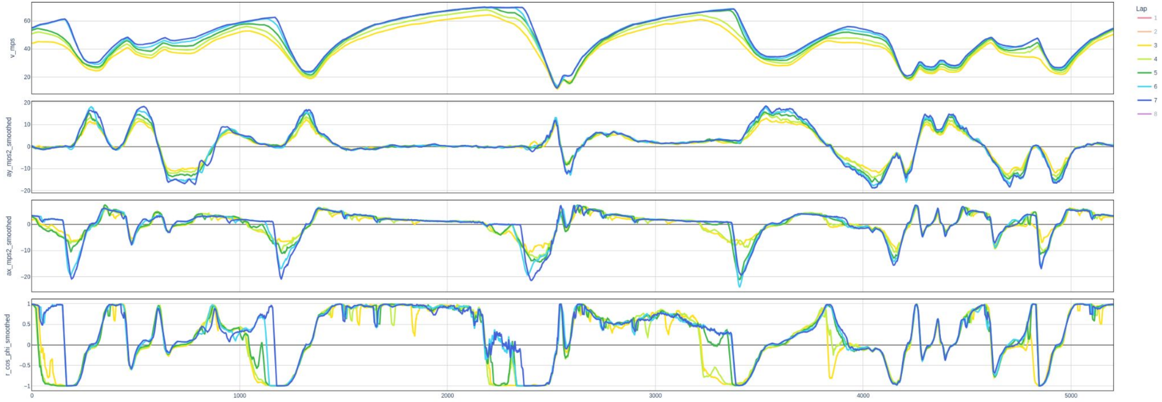


Overlay

After selecting a valid input file, all available channels will be selectable in the in the Overlay Channels tab.

Selected channels will be plotted against the s coordinate (xaxis). Per default velocity, lateral / longitudinal acceleration and rCosPhi will be selected.



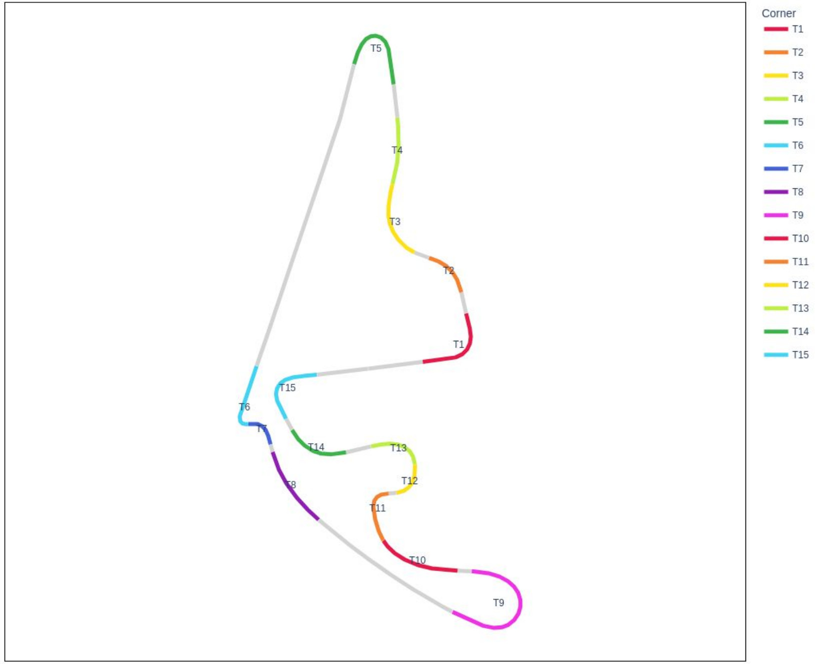


Example plot with default channels and data grouped by lap.

Track Map

Track map colouring is done by different group by options.

Most useful is possibly grouping by corner name to quickly check corner characterisation.



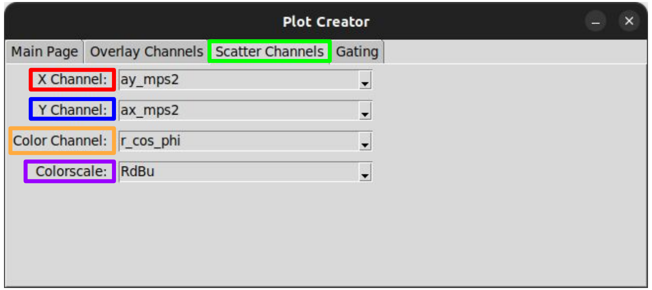
Example plot grouped by corner name and gated for a single lap.

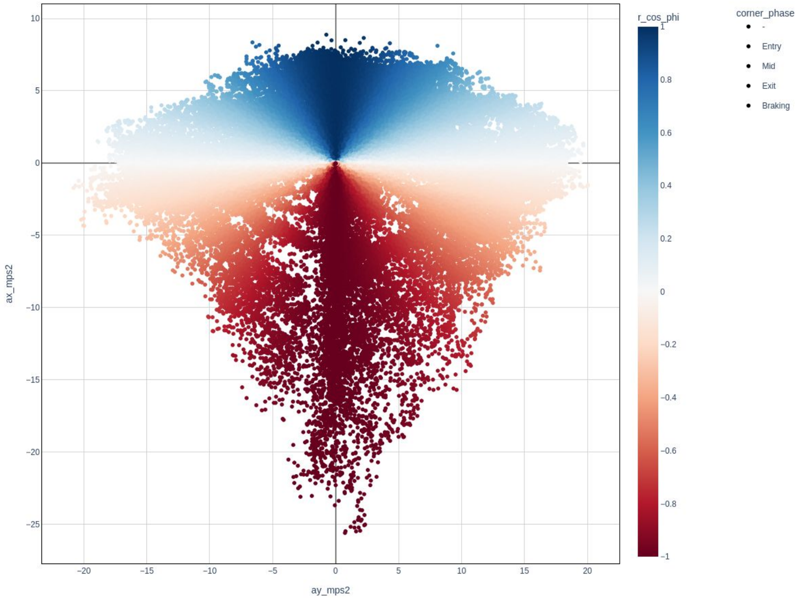
Scatter

1. In order to create a scatter plot x and y channel needs to be selected in the Scatter Channels tab. All available channel options will be selectable (numeric only).

2. There is the option to color the scatter based on a third numeric channel. When None is selected coloring will be based on selected group by option. Selected group by option will still be visible in the legend for changing visibility of data in plot.

3. Colorscale can be changed by the respective dropdown menu.





Example plot showing gg-diagram coloured by rCosPhi.

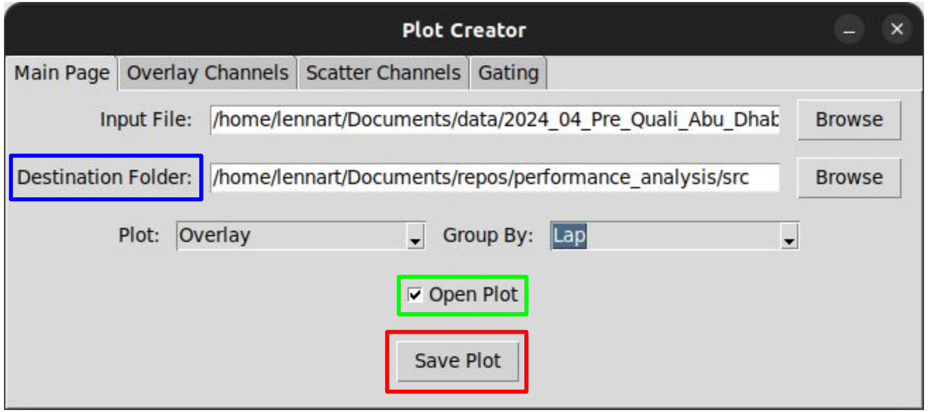
Create Plot

After selecting plotting options, plot can be create by clicking the Save Plot button on the Main Page.

Plot will be saved at the location specified in the Destination Folder field.

If the Open Plot option is checked plot will be opened in the browser.

Plots are created using the plotly library and saved in .html format, which allows for more interactivity.



https://plotly.com/python/